A computable-from-the-halting-set barrier whose free sets compute the halting set, showing tightness of the avoidance theorem

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Editorial note. The counted unit is the article's own unlabelled Proposition at source lines 1262-1264 together with its proof at source lines 1265-1269, a straight adjacent pairing in the pinned file. It is the proposition the article states immediately after its main theorem on strong cone avoidance, and the surrounding prose presents it as the tightness of that theorem. The pinned file is the article's fixed version arXiv:2412.11598v3 under CC BY 4.0.

Editorial note. This statement carries no source label, so the lane identifiers it by its opening line: the claim key is the article's identifier with the addition of the statement's first line, and the statement span is the complete proposition environment from its opening marker to its closing marker. The same convention the project applies to unlabelled statements elsewhere.

Editorial note. Because the statement has no label, the shared gate refuses the interval by design, so the lane ran its documented substitute checks instead: the live occupancy index reports the interval free, no card of any lane carries a token derived from the statement, and the public index carries exactly one entry for this source, at lines 585-587, which does not overlap this interval. The substitution and its outcome are recorded in the card's dedup block and in the lane STATUS.

Editorial note. The retained statement and proof contain no reference to another labelled result and no citation, so this card carries no navigation pointer and no bibliography entry. The notions it uses, the modulus of the halting set and the defining properties of a bounded progressive colouring, are recorded in the standard prerequisites; both are introduced earlier in the article.

Editorial note. The retained preamble is the lane's shared slice for this source, 137 lines.

Allowed standard prerequisites (editorial).

- Bounded progressive colourings, the halting set and its modulus. A colouring into finite sets of natural numbers is progressive when every nonempty

value has least element at least the least element of its argument, and it is constrained by a bounding function when the size of each value is bounded by that function at the least element of the argument; the halting set is the first Turing jump of the empty set and it is equivalent to a total function, its modulus, in the sense that a function dominating that modulus computes the halting set. The author's own notation is retained inside the counted statement and proof.. The counted proposition is the tightness half of the article's main theorem: it exhibits a barrier whose defining colouring is computable from the halting set and whose free sets recover the halting set, so the avoidance result for computable barriers cannot be extended to this one.

Proposition 1. *There exists a $\emptyset'$-computable function $b : \mathbb{N} \to \mathbb{N}$ and a $\emptyset'$-computable b-constrained progressive function $f : \mathbb{N} \to [\mathbb{N}]^{<\omega}$ such that every infinite f-free set computes $\emptyset'$.*

Proof. Let $b : \mathbb{N} \to \mathbb{N}$ be the modulus of $\emptyset'$, and let $f(x) = [x + 1, \ldots, b(x)]$. Let H be an infinite f-free set. Then given $x < y \in H$, $y > b(x)$, so one can H-compute a function dominating b, hence H-compute $\emptyset'$. □